

Arithmetic Masterclass: Percentages, Profit & Loss

Concept Clearance, Advanced Shortcuts, and 20 High-Priority Interview Questions

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Section 1: The Examiner's Blueprint - Core Concepts & Formulas

To clear the TCS NQT Advanced Section, speed and structural clarity are critical. TCS heavily tests base-shifting, rapid fraction-to-percentage conversions, and multi-layered multiplying factors. Relying on textbook variable methods (x) will cause you to run out of time.

1. Fraction-to-Percentage Equivalents Cheat Sheet

Memorizing these fractional values transforms division into instantaneous mental subtraction or multiplication.

Fraction	Percentage	Fraction	Percentage	Fraction	Percentage
1/2	50%	1/6	16.67% / 16 $\frac{2}{3}$ %	1/11	9.09%
1/3	33.33% / 33 $\frac{1}{3}$ %	1/7	14.28% / 14 $\frac{27}{100}$ %	1/12	8.33%
1/4	25%	1/8	12.5% / 12 $\frac{1}{2}$ %	1/15	6.67%
1/5	20%	1/9	11.11% / 11 $\frac{19}{100}$ %	1/20	5%

2. The Multiplying Factor (MF) Technique

Instead of calculating percentage changes as independent steps, look at percentage modification as a single operational multiplier:

Core Multipliers:

- An increase of $x\%$ yields a Multiplying Factor of: $MF = 1 + (x/100)$
- A decrease of $d\%$ (or a discount) yields a Multiplying Factor of: $MF = 1 - (d/100)$

Example: A price increase of 25% corresponds to $MF = 1.25$ (or $5/4$). A discount of 15% corresponds to $MF = 0.85$ (or $17/20$).

3. Successive Percentage Change Shortcut

When an entity undergoes two consecutive percentage changes of **A%** and **B%**, the net percentage change is given by the classic formula:

$$\text{Net Change \%} = A + B + \frac{1}{100}(AB)$$

Sign Convention Rule: Use a positive sign (+) for increases/profits, and a negative sign (-) for decreases/losses/discounts.

4. Profit, Loss, Marked Price & Discount Relationships

TCS questions frequently entangle Cost Price (**CP**), Selling Price (**SP**), and Marked Price (**MP**). Understand how they transition:

$$SP = CP \times [1 + (\text{Profit\%} / 100)] = MP \times [1 - (\text{Discount\%} / 100)]$$

The single most powerful formula to bypass intermediate calculations when dealing with markup and discounts is the **Golden Ratio Link**:

The Golden Ratio Equation:

$$MP / CP = (100 + \text{Profit\%}) / (100 - \text{Discount\%})$$

Section 2: The Examiner's Trap Door - Where TCS Catches You

As test designers, we structure options based on expected student errors. If you fall into these structural traps, you will find your incorrect answer sitting perfectly as Option A or B.

TRAP 1: THE BASE SHIFT FALLACY (PROFIT ON SP VS. CP)

By default, profit is calculated on Cost Price (**CP**). However, TCS love to inject phrases like: "*A merchant calculates his profit percentage on the selling price and finds it to be 20%.*" If a problem states profit on **SP** is 20%, it means **Profit** = $\frac{1}{5} \times SP$. Thus, if **SP** = 5 and **Profit** = 1, the true **CP** = $5 - 1 = 4$. The **true profit percentage** is $(\frac{1}{4}) \times 100 = 25\%$. Always read the denominator reference carefully!

TRAP 2: FABRIC META-TRAPS ("GAINS THE SELLING PRICE OF X ITEMS")

Consider the question: "*By selling 33 meters of cloth, a man gains the selling price of 11 meters.*" Students instantly pick 33.33% ($\frac{11}{33}$). This is wrong!

Write down the basic equation: **Profit** = **SP** - **CP**. Here, $SP(11) = SP(33) - CP(33) \rightarrow CP(33) = SP(22)$. The real base is the cost price of 33 meters, which equals the selling price of 22 meters. The true profit is $\frac{11}{22} \times 100 = 50\%$.

TRAP 3: DISHONEST DEALER FALSE-WEIGHT REFERENCE SHIFTING

When a shopkeeper uses a 800g weight instead of 1kg, students often think the profit is 20% because 200g is saved from 1000g.

Correction: The shopkeeper only invests the cost of 800g from his warehouse to generate the sale. The profit base is always the *actual weight delivered*.

$$\text{Profit\%} = [\text{Error} / (\text{True Value} - \text{Error})] \times 100 = (200 / 800) \times 100 = 25\%.$$

TRAP 4: MULTI-ITEM CAMPAIGN ILLUSIONS ("BUY 3, GET 2 FREE")

Marketing advertisements trap candidates who think the discount is 66.67% or 33.33%. The customer leaves the shop with $3 + 2 = 5$ items. The inventory reduction from the store is 5 items, and the number of un-invoiced items is 2. The true effective discount is $(\text{Free Articles} / \text{Total Articles Received}) \times 100 = (2 / 5) \times 100 = 40\%$.

TRAP 5: CONSTRAINED EQUATIONS WITH EQUAL SP

When two distinct items are sold at identical selling prices, one at $x\%$ profit and the other at $x\%$ loss, there is **always an overall net loss** equal to $(x/10)^2\%$. Do not waste minutes computing absolute costs unless the options are specified in absolute rupee values!

Section 3: 20 High-Priority Exam Questions & Solutions

Question 1: Successive Price and Consumption Base Balancing

Due to a supply shortage, the price of premium sugar increases by 20%. By what percentage should a household reduce its monthly consumption so that the overall expenditure on sugar remains completely unchanged?

- A) 20% B) 16.67%
- C) 15% D) 25%

SOLUTION EXPLANATION:

We know that $\text{Price} \times \text{Consumption} = \text{Expenditure}$. Let initial values be $1 \times 1 = 1$.

New Price Multiplier = **1.20**. Let the new Consumption multiplier be C .

To keep expenditure at 1: $1.20 \times C = 1 \rightarrow C = 1 / 1.20 = 5 / 6$.

The consumption reduction required is $1 - 5/6 = 1/6$. Converting this fraction to percentage: $(1/6) \times 100 = 16.67\%$.

Shortcut: For a constant product, an increase of $x\%$ requires a reduction of $[x / (100 + x)] \times 100 = [20 / 120] \times 100 = 16.67\%$.

Correct Answer: B

Question 2: Multi-Tiered Election Validation Model

In a municipal election between two candidates, 10% of the enrolled voters did not cast their ballots, and 10% of the total votes polled were declared invalid. The winning candidate secured 54% of the valid votes and won by a decisive margin of 1,620 votes. Find the total number of voters enrolled on the voter registration list.

- A) 25,000
B) 20,000
C) 22,500
D) 18,000

SOLUTION EXPLANATION:

Let the total number of registered voters be V .

Votes Polled = $0.90V$.

Valid Votes = $0.90 \times (0.90V) = 0.81V$.

The winner received 54% of the valid votes, which implies the runner-up received $100\% - 54\% = 46\%$ of the valid votes.

The winning margin percentage is $54\% - 46\% = 8\%$ of the valid votes.

Set up the operational equation: $0.08 \times 0.81V = 1620$

$0.0648V = 1620 \rightarrow V = 1620 / 0.0648 = 25,000$.

Correct Answer: A

Question 3: Compound Remainder Expenditure Flow

An engineer spends 25% of his monthly salary on apartment rent, 20% of the remaining salary on food provisions, and 15% of the newly remaining balance on travel expenses. If his final remaining monthly savings equal Rs. 15,300, determine his total monthly gross salary.

- A) Rs. 35,000
B) Rs. 32,000
C) Rs. 30,000
D) Rs. 28,000

SOLUTION EXPLANATION:

Apply successive multiplying factors dynamically instead of setting up intermediate calculations.

Let gross salary be S .

Fraction left after rent = $1 - 0.25 = 0.75$ (or $3/4$).

Fraction left after food = $1 - 0.20 = 0.80$ (or $4/5$).

Fraction left after travel = $1 - 0.15 = 0.85$ (or $17/20$).

Formulate the compound string: $S \times (3/4) \times (4/5) \times (17/20) = 15300$

Notice the terms simplify gracefully: $S \times (3/5) \times (17/20) = 15300 \rightarrow S \times (51/100) = 15300$

$51S = 1,530,000 \rightarrow S = 30,000$.

Shortcut: Notice that $153 / 51 = 3$, meaning the values are chosen by examiners to cancel out instantly!

Correct Answer: C

Question 4: Bi-Directional Demographics Shifts

The population of an industrial township increases by 15% during the first year but experiences a structural contraction of 10% in the second year due to an automation shift. If the current updated population is recorded at 20,700, what was the population of the town exactly two years prior?

- | | |
|-----------|-----------|
| A) 19,000 | B) 20,000 |
| C) 18,500 | D) 21,500 |

SOLUTION EXPLANATION:

Let the population two years ago be P .

Year 1 factor = +15% $\rightarrow 1.15$

Year 2 factor = -10% $\rightarrow 0.90$

Compounded expression: $P \times 1.15 \times 0.90 = 20700$

Evaluate the product: $1.15 \times 0.90 = 1.035$

Thus, $1.035P = 20700 \rightarrow P = 20700 / 1.035 = 20,000$.

Correct Answer: B

Question 5: Two-Dimensional Product Variation Matrix

The length of a commercial rectangular development site is expanded by 30% while its lateral breadth is restricted and shortened by 20%. Calculate the absolute net percentage change manifesting in the overall area of the plot.

- | | |
|-----------------|-----------------------------|
| A) 10% Increase | B) 4% Decrease |
| C) 4% Increase | D) No Net Structural Change |

SOLUTION EXPLANATION:

Area is a product of two independent linear dimensions: $\text{Area} = \text{Length} \times \text{Breadth}$.

Apply the successive percentage change formula directly: $\text{Net Change} = A + B + (AB / 100)$

Substitute the parameters using correct sign conventions ($A = +30$, $B = -20$):

$\text{Net Change} = 30 + (-20) + [(30 \times -20) / 100] = 10 + (-600 / 100) = 10 - 6 = +4\%$.

The positive output denotes an absolute increase of 4%.

Correct Answer: C

Question 6: Constrained Equal Selling Price Dual-Transaction Model

A high-volume dealer disposes of two electronic units for exactly Rs. 9,900 each. On the first transaction, he realizes a clean gain of 10%, while on the second transaction, he sustains a loss of 10%. Determine his net aggregate profit or loss percentage across the cumulative process.

A) No Profit, No Loss

B) 1% Gain

C) 2% Loss

D)
1%
Loss

SOLUTION EXPLANATION:

TCS expects you to calculate the individual cost prices ($CP_1 = 9900 / 1.1 = 9000$ and $CP_2 = 9900 / 0.9 = 11000$, total $CP = 20000$; total $SP = 19800$, loss = 200, which is 1%). Do not do this unless an absolute rupee figure is requested.

Examiner's Rule: When $SP_1 = SP_2$, and the profit percentage on one equals the loss percentage on the other ($x\%$), the system encounters a perpetual net loss calculated via:

$$Loss \% = (x / 10)^2 = (10 / 10)^2 = 1^2 = 1\%$$

Correct Answer: D

Question 7: Dishonest Shopkeeper Fraudulent Mass Manipulation

A grocery vendor claims to offer his commodities at absolute factory cost price but substitutes a fraudulent calibration scale that delivers 920 grams in place of a standard 1 kilogram mass package. What is his exact actual yield profit percentage?

A) 8.00%

B) 8.69%

C) 9.20%

D) 11.11%

SOLUTION EXPLANATION:

The vendor's real investment is only the cost value of the 920g exiting his inventory store, while his revenue cash flow matches a 1000g valuation.

Absolute mass saved (Error) = $1000g - 920g = 80g$.

The calculation base is the delivered weight (920g):

$Profit \% = [Error / True Weight Delivered] \times 100 = [80 / 920] \times 100 = [2 / 23] \times 100 = 200 / 23 \approx 8.69\%$.

Correct Answer: B

Question 12: Fixed Asset Cost-Revenue Equivalence Matrix

An operations audit establishes that the production Cost Price of 15 standard corporate laptop units matches the outward retail Selling Price realized from exactly 12 laptop units. Determine the real profit rate generated.

- A) 25.00% B) 20.00%
- C) 33.33% D) 16.67%

SOLUTION EXPLANATION:

Translate the statement directly into a functional algebraic equation:

$$15 \times CP = 12 \times SP$$

Isolate the structural cost-revenue ratio: $SP / CP = 15 / 12$.

Reducing the fraction to lowest terms gives: $SP / CP = 5 / 4$.

This indicates that a capital input of 4 units generates a revenue output of 5 units, establishing a clean profit margin of $5 - 4 = 1$ unit.

$$\text{Profit rate} = (1 / 4) \times 100 = 25\%.$$

Correct Answer: A

Question 13: Linear Cascading Supply-Chain Valuations

A primary component manufacturer ships a finished batch to an industrial distributor at a clean 10% profit. The distributor forwards the batch to a local retailer securing a 20% margin enhancement. The retailer finally completes the consumer sale at a 25% markup. If the end consumer pays an invoice total of Rs. 1,650, find the manufacturer's base cost of production.

- A) Rs. 1,200 B) Rs. 1,100
- C) Rs. 1,000 D) Rs. 950

SOLUTION EXPLANATION:

Let the original manufacturing base cost be M .

Chain the independent multi-tier compounding factors:

$$M \times 1.10 \times 1.20 \times 1.25 = 1650$$

Group the operations to expedite computation. Notice that $1.20 \times 1.25 = 5/4 \times 6/5 = 1.50$.

$$\text{The calculation reduces down to: } M \times 1.10 \times 1.50 = 1650 \rightarrow M \times 1.65 = 1650.$$

$$\text{Solving for the base production cost: } M = 1650 / 1.65 = 1,000.$$

Correct Answer: C

Question 16: Interlocking Discount and Fiscal Tax Operations

A consumer electronics boutique offers an upfront promotional discount of 15% on the catalog marked price of a device, but applies a mandatory 8% value-added municipal tax on the actual discounted checkout price. If a shopper pays a total checkout invoice of Rs. 22,950, what was the original raw catalog marked price?

- A) Rs. 26,000 B) Rs. 25,000
C) Rs. 24,000 D) Rs. 23,500

SOLUTION EXPLANATION:

Let the original catalog Marked Price be **MP**.

Apply the discount multiplier factor: **Post-Discount Price** = **MP** × **0.85**.

Apply the fiscal tax multiplier factor on this intermediate value: **Final Check-out Price** = (**MP** × **0.85**) × **1.08**.

Evaluate the combined mathematical multiplier: **0.85** × **1.08** = **0.918**.

Equate to the target invoice amount: **0.918** × **MP** = **22950**

MP = **22950 / 0.918** = **25,000**.

Correct Answer: B

Question 17: Released Capital Arbitrage and Volume Expansion

Due to a global production surge, the wholesale price of premium coffee beans drops by 25%. This drop allows a roasting firm to purchase an additional 4 kilograms of coffee for the same fixed budget allocation of Rs. 1,200. Calculate the updated reduced price per kilogram.

- A) Rs. 100/kg B) Rs. 80/kg
C) Rs. 75/kg D) Rs. 60/kg

SOLUTION EXPLANATION:

Examiner's Fast-Track: Think about the cash interaction. A price drop of 25% means that for the same volume, the client would save exactly 25% of his budget.

Released Capital Savings = 25% of Rs. 1200 = **0.25** × **1200** = **Rs. 300**.

Because the customer chooses to maintain the full budget expenditure, this released sum of Rs. 300 pays for the extra 4 kg of coffee beans.

Therefore, the updated reduced price per kilogram is directly: **Rs. 300 / 4 kg** = **Rs. 75 per kg**.

Correct Answer: C

Question 18: Dual-Component Capital Alligation Cross-Over Model

A transport company acquires a used cargo scooter and an auxiliary distribution bicycle for an aggregate total investment of Rs. 25,000. The scooter is liquidated at a 20% profit, while the bicycle is sold at a 10% loss. If the aggregate return across the entire mixed transaction yields a net gain of 8%, find the independent initial cost price allocation of the scooter.

A) Rs. 16,000

B) Rs. 15,000

C) Rs. 12,500

D) Rs. 10,000

SOLUTION EXPLANATION:

This problem is solved most efficiently by applying the weighted average rule of Alligation.

Left Wing (Scooter Return Rate) = $+20\%$

Right Wing (Bicycle Return Rate) = -10%

Center Hub (Aggregate Net Target Return) = $+8\%$

Compute diagonal absolute variances to get the capital weight ratios:

Scooter weight component = $|8 - (-10)| = 18$

Bicycle weight component = $|20 - 8| = 12$

Ratio of Cost Prices ($CP_{\{Scooter\}} : CP_{\{Bicycle\}}$) = $18 : 12 = 3 : 2$.

Divide the total capital into $3 + 2 = 5$ proportional parts:

$5 \text{ parts} = \text{Rs. } 25,000 \rightarrow 1 \text{ part} = \text{Rs. } 5,000$.

Cost Price allocation of the scooter = $3 \text{ parts} \times \text{Rs. } 5,000 = \text{Rs. } 15,000$.

Correct Answer: B

